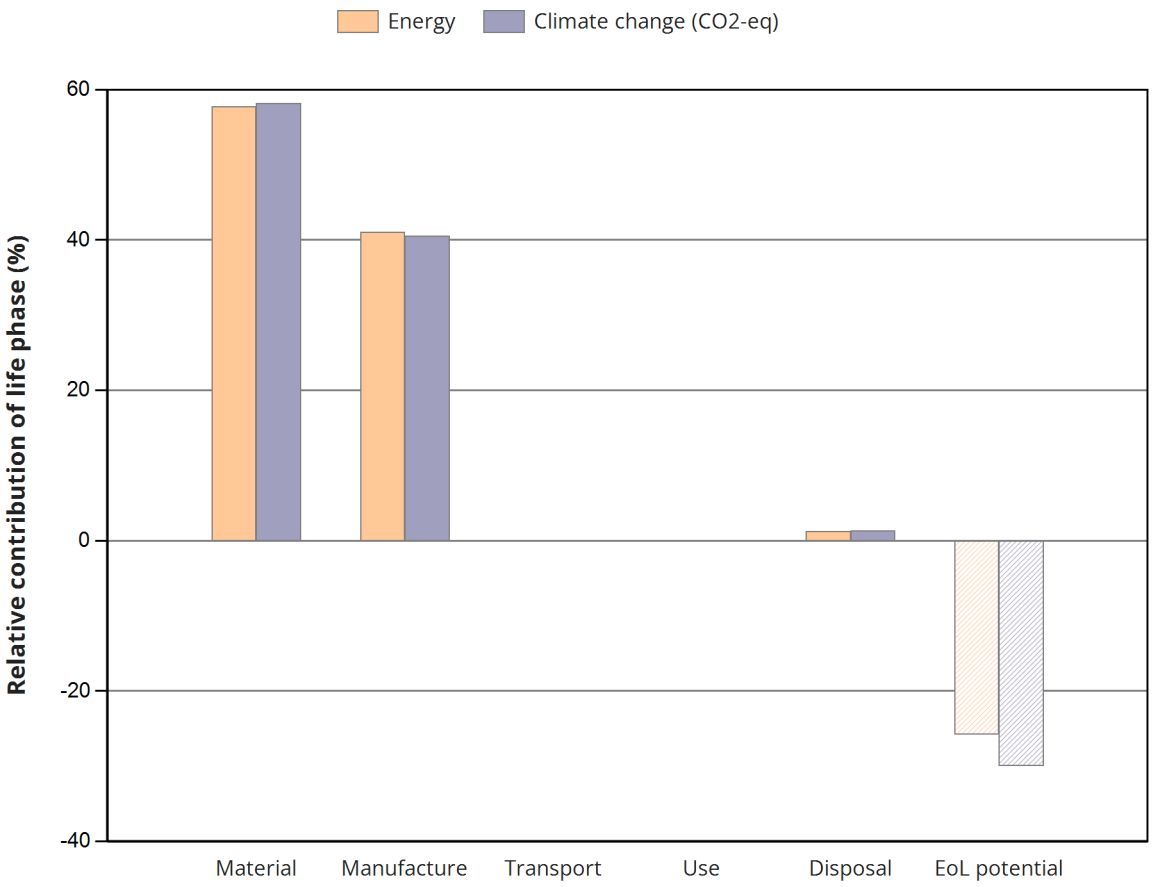
Eco Audit Report

Product name	EIMMORTAL 2026 ATV
Country of use	The Whole World
Product life (years)	5

Summary:



Energy details Climate change (CO2-eq) details

Phase	Energy (MJ)	Energy (%)	Climate change (CO2-eq) (kg)	Climate change (CO2-eq) (%)
Material	1.99e+03	57.7	133	58.2
Manufacture	1.41e+03	41.0	92.6	40.5
Transport	0	0.0	0	0.0
Use	0	0.0	0	0.0
Disposal	42.5	1.2	2.98	1.3
Total (for first life)	**3.45e+03**	**100**	**229**	**100**
End of life potential	-888		-68.4	

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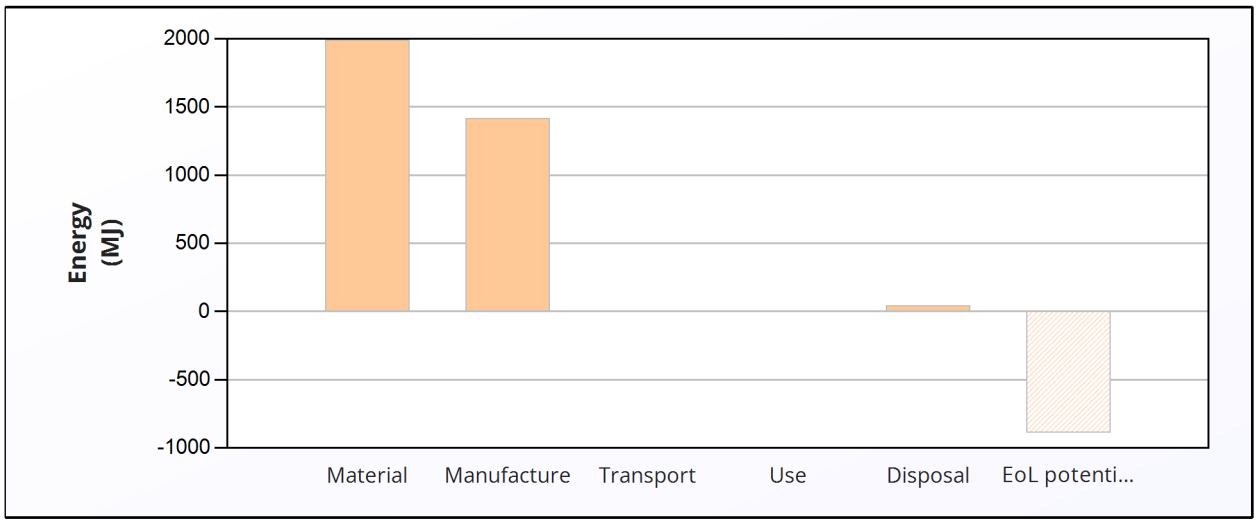
ENERGY ANALYSIS

Summary



	Energy (MJ/year)
Equivalent annual environmental burden (averaged over 5 year product life):	690

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Detailed breakdown of individual life phases

Material:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass processed** (kg)	Energy (MJ)	%
Battery Casing	Aluminum, 6061, T6	Reused part	3	1	3	0	0.0
Wiring	Copper, C10200, soft (oxygen-free h.c. copper)	28.0%	1.5	1	1.5	90	4.5
Rollcage (Secondary 25.4 mm)	Low alloy steel, AISI 9310, normalized	30.0%	14	1	16	3.8e+02	19.2
Fr. Gearbox	Carbon steel, AISI 1040, annealed	Virgin (0%)	3.8	1	9.4	1.8e+02	8.9
Rr. Gearbox	Carbon steel, AISI 1040, annealed	Virgin (0%)	6.5	1	16	3.1e+02	15.4
Driveshaft	Carbon steel, AISI 1040, annealed	Reused part	13	1	13	0	0.0
Uprights	Carbon steel, AISI 1080, annealed	Virgin (0%)	6	1	11	2.4e+02	12.1
Suspension Arms	Low alloy steel, AISI 4130, annealed	Virgin (0%)	9	1	10	3e+02	15.0
Body Panel	PC+Polyester transparent amorphous (optical quality)	Virgin (0%)	3.4	1	3.4	3.7e+02	18.5
CVT Casing	Carbon steel, AISI 1080, annealed	Virgin (0%)	3	1	6	1.3e+02	6.3
Seat	Polyurethane foam (elastomeric, open cell, 0.032)	Reused part	2.8	1	2.8	0	0.0
Total				**11**	**93**	**2e+03**	**100**

*Typical: Includes 'recycle fraction in current supply'

**Where applicable, includes material mass removed by secondary processes

***User-defined material

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MANUFACTURE:

Summary

Component	Process	% Removed	Amount processed	Energy (MJ)	%
Wiring	Wire drawing	-	1.5 kg	21	1.5
Rollcage (Secondary 25.4 mm)	Metal rolling and forging	-	16 kg	2.5e+02	18.0
Rollcage (Secondary 25.4 mm)	Cutting and trimming	11	1.8 kg	0.53	0.0
Fr. Gearbox	Metal rolling and forging	-	9.4 kg	95	6.7
Fr. Gearbox	Machining, fine	60	5.6 kg	1.3e+02	9.4
Rr. Gearbox	Metal rolling and forging	-	16 kg	1.6e+02	11.6
Rr. Gearbox	Machining, fine	60	9.8 kg	2.3e+02	16.2
Uprights	Metal rolling and forging	-	11 kg	1.2e+02	8.2
Uprights	Machining, fine	45	4.9 kg	1.2e+02	8.4
Suspension Arms	Metal rolling and forging	-	10 kg	1.2e+02	8.3
Suspension Arms	Cutting and trimming	11	1.1 kg	0.33	0.0
Body Panel	Polymer extrusion	-	3.4 kg	29	2.0
Body Panel	Cutting and trimming	-	0 kg	0	0.0
CVT Casing	Metal rolling and forging	-	6 kg	64	4.5
CVT Casing	Machining, fine	50	3 kg	72	5.1
Total				**1.4e+03**	**100**

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Transport:

Summary

Breakdown by transport stage

Stage name	Transport type	Distance (km)	Energy (MJ)	%
Total				100

Breakdown by components

Component	Mass (kg)	Energy (MJ)	%
Battery Casing	3	0	
Wiring	1.5	0	
Rollcage (Secondary 25.4 mm)	14	0	
Fr. Gearbox	3.8	0	
Rr. Gearbox	6.5	0	
Driveshaft	13	0	
Uprights	6	0	
Suspension Arms	9	0	
Body Panel	3.4	0	
CVT Casing	3	0	
Seat	2.8	0	
Total	**66**	**0**	**100**

Use:

Summary

Relative contribution of static and mobile modes

Mode	Energy (MJ)	%
Static	0	
Mobile	0	
Total	**0**	**100**

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Disposal:

Summary

Component	End of life option	% recovered	Energy (MJ)	%
Battery Casing	Recycle	90.0	2	4.6
Wiring	Recycle	100.0	1.1	2.5
Rollcage (Secondary 25.4 mm)	Recycle	98.0	9.9	23.4
Fr. Gearbox	Recycle	98.0	2.6	6.1
Rr. Gearbox	Recycle	98.0	4.5	10.5
Driveshaft	Recycle	97.0	8.9	20.9
Uprights	Recycle	95.0	4.1	9.5
Suspension Arms	Recycle	98.0	6.2	14.6
Body Panel	Landfill	0.0	0.68	1.6
CVT Casing	Recycle	100.0	2.1	4.9
Seat	Landfill	0.0	0.56	1.3
Total			**43**	**100**

EoL potential:

Component	End of life option	% recovered	Energy (MJ)	%
Battery Casing	Recycle	90.0	0	0.0
Wiring	Recycle	100.0	-62	7.0
Rollcage (Secondary 25.4 mm)	Recycle	98.0	-2.2e+02	24.5
Fr. Gearbox	Recycle	98.0	-81	9.1
Rr. Gearbox	Recycle	98.0	-1.4e+02	15.8
Driveshaft	Recycle	97.0	0	0.0
Uprights	Recycle	95.0	-1.3e+02	14.2
Suspension Arms	Recycle	98.0	-1.9e+02	21.9
Body Panel	Landfill	0.0	0	0.0
CVT Casing	Recycle	100.0	-66	7.4
Seat	Landfill	0.0	0	0.0
Total			**-8.9e+02**	**100**

Notes:

Summary

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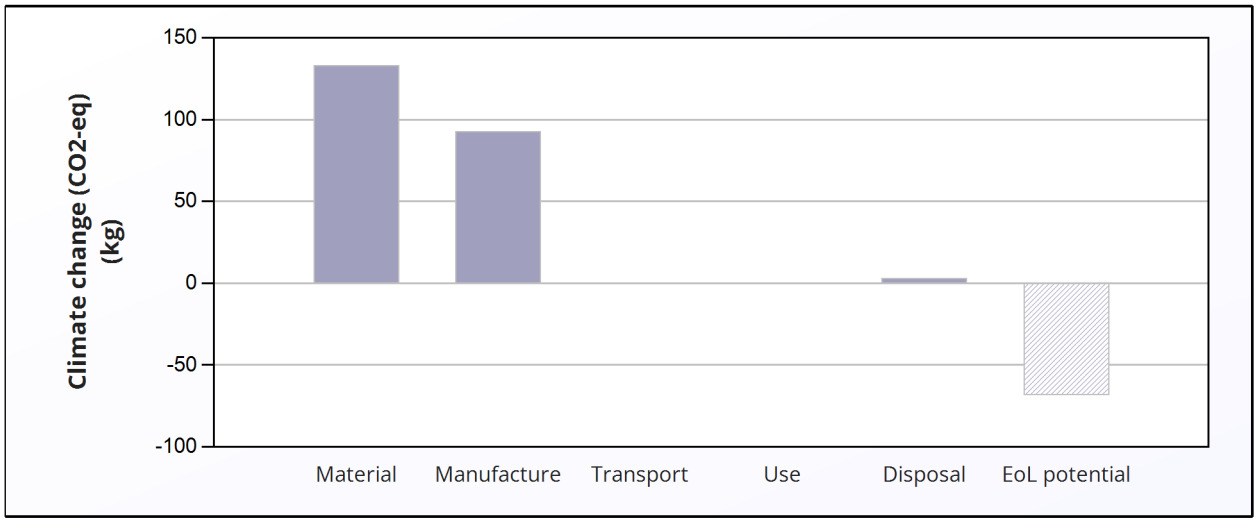
CLIMATE CHANGE (CO2-EQ) ANALYSIS

Summary



	CO2 (kg/year)
Equivalent annual environmental burden (averaged over 5 year product life):	45.7

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Detailed breakdown of individual life phases

Material:

Summary

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass processed** (kg)	Climate change (CO2-eq) (kg)	%
Battery Casing	Aluminum, 6061, T6	Reused part	3	1	3	0	0.0
Wiring	Copper, C10200, soft (oxygen-free h.c. copper)	28.0%	1.5	1	1.5	5.9	4.5
Rollcage (Secondary 25.4 mm)	Low alloy steel, AISI 9310, normalized	30.0%	14	1	16	28	20.7
Fr. Gearbox	Carbon steel, AISI 1040, annealed	Virgin (0%)	3.8	1	9.4	12	9.4
Rr. Gearbox	Carbon steel, AISI 1040, annealed	Virgin (0%)	6.5	1	16	22	16.2
Driveshaft	Carbon steel, AISI 1040, annealed	Reused part	13	1	13	0	0.0
Uprights	Carbon steel, AISI 1080, annealed	Virgin (0%)	6	1	11	17	13.0
Suspension Arms	Low alloy steel, AISI 4130, annealed	Virgin (0%)	9	1	10	22	16.4
Body Panel	PC+Polyester transparent amorphous (optical quality)	Virgin (0%)	3.4	1	3.4	17	13.1
CVT Casing	Carbon steel, AISI 1080, annealed	Virgin (0%)	3	1	6	9	6.8
Seat	Polyurethane foam (elastomeric, open cell, 0.032)	Reused part	2.8	1	2.8	0	0.0
Total				**11**	**93**	**1.3e+02**	**100**

*Typical: Includes 'recycle fraction in current supply'

**Where applicable, includes material mass removed by secondary processes

***User-defined material

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MANUFACTURE:

Summary

Component	Process	% Removed	Amount processed	Climate change (CO2-eq) (kg)	%
Wiring	Wire drawing	-	1.5 kg	1.4	1.5
Rollcage (Secondary 25.4 mm)	Metal rolling and forging	-	16 kg	17	18.0
Rollcage (Secondary 25.4 mm)	Cutting and trimming	11	1.8 kg	0.041	0.0
Fr. Gearbox	Metal rolling and forging	-	9.4 kg	6.2	6.7
Fr. Gearbox	Machining, fine	60	5.6 kg	8.7	9.4
Rr. Gearbox	Metal rolling and forging	-	16 kg	11	11.6
Rr. Gearbox	Machining, fine	60	9.8 kg	15	16.2
Uprights	Metal rolling and forging	-	11 kg	7.6	8.2
Uprights	Machining, fine	45	4.9 kg	7.7	8.4
Suspension Arms	Metal rolling and forging	-	10 kg	7.7	8.3
Suspension Arms	Cutting and trimming	11	1.1 kg	0.026	0.0
Body Panel	Polymer extrusion	-	3.4 kg	1.9	2.0
Body Panel	Cutting and trimming	-	0 kg	0	0.0
CVT Casing	Metal rolling and forging	-	6 kg	4.2	4.5
CVT Casing	Machining, fine	50	3 kg	4.7	5.1
Total				**93**	**100**

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Transport:

Summary

Breakdown by transport stage

Stage name	Transport type	Distance (km)	Climate change (CO2-eq) (kg)	%
Total				100

Breakdown by components

Component	Mass (kg)	Climate change (CO2-eq) (kg)	%
Battery Casing	3	0	
Wiring	1.5	0	
Rollcage (Secondary 25.4 mm)	14	0	
Fr. Gearbox	3.8	0	
Rr. Gearbox	6.5	0	
Driveshaft	13	0	
Uprights	6	0	
Suspension Arms	9	0	
Body Panel	3.4	0	
CVT Casing	3	0	
Seat	2.8	0	
Total	**66**	**0**	**100**

Use:

Summary

Relative contribution of static and mobile modes

Mode	Climate change (CO2-eq) (kg)	%
Static	0	
Mobile	0	
Total	**0**	**100**

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Disposal:

Summary

Component	End of life option	% recovered	Climate change (CO2-eq) (kg)	%
Battery Casing	Recycle	90.0	0.14	4.6
Wiring	Recycle	100.0	0.074	2.5
Rollcage (Secondary 25.4 mm)	Recycle	98.0	0.7	23.4
Fr. Gearbox	Recycle	98.0	0.18	6.1
Rr. Gearbox	Recycle	98.0	0.31	10.5
Driveshaft	Recycle	97.0	0.62	20.9
Uprights	Recycle	95.0	0.28	9.5
Suspension Arms	Recycle	98.0	0.43	14.6
Body Panel	Landfill	0.0	0.048	1.6
CVT Casing	Recycle	100.0	0.15	4.9
Seat	Landfill	0.0	0.039	1.3
Total			**3**	**100**

EoL potential:

Component	End of life option	% recovered	Climate change (CO2-eq) (kg)	%
Battery Casing	Recycle	90.0	0	0.0
Wiring	Recycle	100.0	-4.3	6.2
Rollcage (Secondary 25.4 mm)	Recycle	98.0	-17	24.7
Fr. Gearbox	Recycle	98.0	-6.3	9.2
Rr. Gearbox	Recycle	98.0	-11	16.0
Driveshaft	Recycle	97.0	0	0.0
Uprights	Recycle	95.0	-9.8	14.3
Suspension Arms	Recycle	98.0	-15	22.1
Body Panel	Landfill	0.0	0	0.0
CVT Casing	Recycle	100.0	-5.1	7.5
Seat	Landfill	0.0	0	0.0
Total			**-68**	**100**

Notes:

Summary

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